

MATH 1200 TUTORIAL 8: SETS AND FUNCTIONS

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1. SETS

At this point, we should be familiar with some sets, such as

- $\mathbb{N} = \{1, 2, 3, \dots\}$: the natural numbers.
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$: the integers.
- $\mathbb{Q} = \{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$: the rational numbers.
- $\mathbb{R}$ the real numbers.

However, there are times where we would like to consider an arbitrary set, and some sets in which conditions are provided.

Definition 1.1: A *set* is an unordered collection of disjoint objects, called *elements* or *members* of the set. A set is said to *contain* its elements. We write $x \in A$ to denote that x is an element of the set A . The notation $x \notin A$ denotes that x is not a member of A .

Example 1.2: The following are examples are sets:

- The set V of all vowels in the English alphabet can be written as $V = \{a, e, i, o, u\}$.
- The set O of positive integers less than 10 can be expressed as $O = \{1, 3, 5, 7, 9\}$.
- The set of positive integers less than 100 can be denoted by $\{1, 2, \dots, 99\}$.

The sets that are described in Example 1.2 is known as the *roster method*. Another way to describe a set is to use *set builder* notation. We characterize those elements in the set by stating the property or properties they must have to be members. In general, the *set builder* notation is of the form

$$\{x : x \text{ has property } P\}$$

Example 1.3: Considering the examples in Example 1.2

- If $\mathbb{A}$ denotes the set of all letters in the English alphabet, then

$$V = \{x \in \mathbb{A} : x \text{ is a vowel}\}$$

- The set of all odd positive integers can be written as

$$O = \{x : x \text{ is a positive integer less than } 10\} = \{x \in \mathbb{N} : x \text{ is odd and } x < 10\}$$

Definition 1.4: Let A and B be sets.

- We say that A is a *subset* of B , and denote $A \subset B$, if every element in A is in B . That is, if $x \in A$, then $x \in B$.
- We say that A and B are *equal*, and denote $A = B$, if $A \subset B$ and $B \subset A$.
- We say that A is a *proper subset* of B , and denote $A \subsetneq B$, if x is an element of B but not an element of A .

Example 1.5: Let A , B , and C be sets. Show that if $A \subset B$ and $B \subset C$, then $A \subset C$.

Let $x \in A$. Because $A \subset B$, then we have that $x \in B$. Furthermore, because $B \subset C$, then we have $x \in C$. Therefore, because $x \in A$ implies $x \in C$, then $A \subset C$, as required.

Example 1.6: Prove that $\overset{A}{\{(2a - 3b, a, b) : a, b \in \mathbb{R}\}} = \overset{B}{\{(x + 5y, 2x + y, x - y) : x, y \in \mathbb{R}\}}$

To show two sets are equal, we show $A \subset B$ and $B \subset A$.

Let $z \in A$, i.e. $z = (2a - 3b, a, b)$. To show $z \in B$, we need to find some x and y such that

$$2a - 3b = x + 5y$$

$$a = 2x + y$$

$$b = x - y.$$

In particular, $a = 2x + y$ and $b = x - y$, so $z \in B$.

Similar, find x, y to show $z \in B \Rightarrow z \in A$.

We will now have a look at some set operations that we can perform on sets.

Definition 1.7: Let A and B be sets on some universal set X

- (i) The *union of A and B* , denoted by $A \cup B$, is the set of all elements that are in either A or B . In other words,

$$A \cup B = \{x \in X : x \in A \vee x \in B\}$$

- (ii) The *intersection of A and B* , denoted by $A \cap B$, is the set of all elements that are in both A and B . In other words,

$$A \cap B = \{x \in X : x \in A \wedge x \in B\}$$

- (iii) The *complement of A* , denoted by A^c is the set of all elements that are not in A . In other words,

$$A^c = \{x \in X : x \notin A\}$$

- (iv) The *set difference of A and B* , denoted by $A \setminus B$ is the set of all elements that are in A but not in B . In other words,

$$A \setminus B = \{x \in X : x \in A \wedge x \notin B\}$$

Remark 1.8: It is important to note that there is a distinction when saying $A \setminus B$ and $B \setminus A$, which will be a common mistake among students.

Remark 1.9: It is possible to extend the union and intersection to more than two sets. Indeed, if we have sets $A_1, A_2, A_3, \dots, A_n$, then we can write

- The union of $A_1, A_2, \dots, A_n$ as

$$A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i$$

- The intersection of $A_1, A_2, \dots, A_n$ as

$$A_1 \cap A_2 \cap \dots \cap A_n = \bigcap_{i=1}^n A_i$$

Example 1.10 (De Morgan's Law): Prove that

- (i) If A and B are sets, then $\overline{A \cup B} = \bar{A} \cap \bar{B}$. $(A \cup B)^c = A^c \cap B^c$
 (ii) If $A_1, A_2, \dots, A_n$ are sets, $\overline{\bigcup_{i=1}^n A_i} = \bigcap_{i=1}^n \bar{A}_i$.

(i) To show $(A \cup B)^c = A^c \cap B^c$, we show $(A \cup B)^c \subset A^c \cap B^c$ and $A^c \cap B^c \subset (A \cup B)^c$

- If $x \in (A \cup B)^c$, then $x \notin A \cup B$, so $x \notin A$ or $x \notin B$, but $x \in A^c$ and $x \in B^c$, so $x \in A^c \cap B^c$.
- If $x \in A^c \cap B^c$, then $x \in A^c$ and $x \in B^c$. But $x \notin A$ or $x \notin B$, so $x \notin A \cup B$, thus $x \in (A \cup B)^c$.

Example continued...

(ii) To show that $(\bigcup_{i=1}^n A_i)^c = \bigcap_{i=1}^n A_i^c$, we show by double inclusion.

- Let $x \in (\bigcup_{i=1}^n A_i)^c$. Then $x \notin \bigcup_{i=1}^n A_i$, so $x \notin A_1$ or $x \notin A_2$, ..., $x \notin A_n$. So $x \in A_1^c$ and $x \in A_2^c$ and, ..., $x \in A_n^c$,
so $x \in \bigcap_{i=1}^n A_i^c$.
- Let $x \in \bigcap_{i=1}^n A_i^c$. Then $x \in A_1^c$, ..., $x \in A_n^c$, so $x \notin A_1$ or $x \notin A_2$ or ... or $x \notin A_n$, so $x \notin \bigcup_{i=1}^n A_i \Rightarrow x \in (\bigcup_{i=1}^n A_i)^c$.

Exercise 1.11: Prove that

- If A and B are sets, then $\overline{A \cap B} = \bar{A} \cup \bar{B}$.
- If $A_1, A_2, \dots, A_n$ are sets, then $\overline{\bigcap_{i=1}^n A_i} = \bigcup_{i=1}^n \bar{A}_i$.

2. FUNCTIONS

Back in high school, we have learned several kinds of functions including polynomial, trigonometric, exponential, logarithmic, and even composition of functions. When we get to higher level mathematics, we would like to consider functions in a general sense, rather than looking at examples of functions.

The idea is that we would like to relate sets that we have talked about earlier, and apply it to talk about functions.

Definition 2.1: Let A and B be nonempty sets. A *function* from A to B is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element $a \in A$. If f is a function from A to B , we write $f : A \rightarrow B$.

Definition 2.2: Let f be a function from A to B .

- (i) We say that A is the *domain* of f , and B is the *codomain* of f .
- (ii) If $f(a) = b$, we say that b is the *image* of a , and a is the *preimage* of b .
- (iii) The *range* or *image* of f is the set of all images of elements of A .

Example 2.3: If $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is the map that assigns the square of an integer to an integer given by $f(n) = n^2$, what is the domain, codomain, and the range?

Domain : $\mathbb{Z}$ $f(1) = 1, f(2) = 4, f(3) = 9, \dots$
 Codomain : $\mathbb{Z}$
 Range : $\{1, 4, 9, 16, \dots\} = \{n^2 : n \in \mathbb{Z}\}$

Definition 2.4: Let f and g be functions from A to B . Then $f + g$ and fg are also functions from A to B defined for all $x \in A$

- $(f + g)(x) = f(x) + g(x)$
- $(fg)(x) = f(x)g(x)$

Example 2.5: If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are given by $f(x) = x^2$ and $g(x) = x - x^2$, then $(f + g)(x) = x$ and $(fg)(x) = x^3 - x^4$.

Definition 2.6: Let f be a function from A to B and let S be a subset of A . The *image* of S under the function f is the subset of B that consists of the images of the elements of S . We denote the image of S by

$$f[S] = \{y \in B : \text{there exists an } x \in S \text{ such that } f(x) = y\} = \{f(x) : x \in S\}$$

Now that we have the preliminary details of what it means for a map to be a function, we will now discuss four important concepts about functions.

Definition 2.7: Let f be a function from A to B .

- (i) We say that f is *one-to-one*, or an *injection* if and only if for every $x, y \in A$, $f(x) = f(y)$ implies $x = y$.
- (ii) We say that f is *onto* if for every $y \in B$, there exists an $x \in A$ such that $f(x) = y$.
- (iii) We say that f is a *bijection* if f is both one-to-one and onto.

(iv) We say that f has an *inverse* if f is a bijection, and its inverse is defined by the map $f^{-1} : B \rightarrow A$.

Example 2.8: Determine if the following functions are (a) one-to-one, (b) onto, (c) bijective, and (d) if it has an inverse.

(i) $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 2x + 1$

$$y = 2x + 1 \quad x = \frac{y-1}{2}$$

One-to-one: Let $x, y \in \mathbb{R}$ such that $f(x) = f(y)$. Then
 $2x + 1 = 2y + 1 \Rightarrow 2x = 2y \Rightarrow x = y$. (Yes).

Onto: Let $y \in \mathbb{R}$. Find $x \in \mathbb{R}$ so that $f(x) = y$.

$$2x + 1 = y \Rightarrow 2x = y - 1 \Rightarrow x = \frac{y-1}{2} \quad (\text{onto})$$

Bijection: Yes, because one-to-one and onto.

Inverse: $f^{-1}(x) = \frac{x-1}{2}$

(ii) $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2 + 1$

One-to-one: Take $1, -1 \in \mathbb{R}$ such that $f(1) = f(-1)$
 which is true because $1^2 + 1 = (-1)^2 + 1 \Rightarrow 2 = 2$
 but $1 \neq -1$, so not one-to-one.

Onto: There is no $x \in \mathbb{R}$ so that $f(x) = 0$,
 hence, not onto.

Bijection: No. Inverse: Does not exist.

(iii) $f : \mathbb{R} \rightarrow (0, \infty)$ given by $f(x) = e^x$

One-to-one: Let $x, y \in \mathbb{R}$ such that $f(x) = f(y)$.

Then $e^x = e^y$ which implies $x = y$. Hence one-to-one.

Onto: Yes. Because for any $x \in \mathbb{R}$, $f(x) > 0$,
 and our codomain is same as the range, it
 is onto.

Note: If $f(\text{Domain}) = \text{Range}$, onto.

Bijection: Yes

Inverse: $y = e^x \Rightarrow x = e^y \Rightarrow \ln(x) = y \ln(e)$
 $\Rightarrow f^{-1}(x) = \ln(x)$.

Another important concept to keep in mind is the idea of composing functions.

Definition 2.9: Let $f : A \rightarrow B$ and let $g : B \rightarrow C$. The *composition* of f and g , denoted by $g \circ f$ is the function from A to C defined by $(g \circ f)(x) = g(f(x))$.

Example 2.10: Let $f(x) = 2x + 3$ and $g(x) = 3x + 2$. Find $f \circ g$ and $g \circ f$.

$$\begin{aligned} f(g(x)) &= f(3x + 2) = 2(3x + 2) + 3 \\ &= 6x + 4 + 3 = 6x + 7 \end{aligned}$$

$$\begin{aligned} g(f(x)) &= g(2x + 3) = 3(2x + 3) + 2 \\ &= 6x + 9 + 2 \\ &= 6x + 11. \end{aligned}$$

Let us now identify some results about functions and composition of functions.

Example 2.11: If $f : A \rightarrow B$ and $g : B \rightarrow C$, then

$$g \circ f : A \rightarrow C$$

(i) If $g \circ f$ is onto, then g is onto.

If $g \circ f$ is onto, then for any $c \in C$, there exists an $a \in A$ such that $(g \circ f)(a) = g(f(a)) = c$.

To see that g is onto, we seek a $b \in B$ such that $g(b) = c$. In particular, we take $b = f(a)$ so that $g(b) = g(f(a)) = c$, hence g is onto.

(ii) If $g \circ f$ is one-to-one, then f is one-to-one.

Assume that $g \circ f$ is one-to-one. Then we have for any $x, y \in A$,

$$\begin{aligned} (g \circ f)(x) = (g \circ f)(y) &\Rightarrow g(f(x)) = g(f(y)) \\ &\Rightarrow x = y. \end{aligned}$$

For a contradiction, if f is not one-to-one, then there are $z, w \in A$ such that

$$f(z) = f(w) \text{ but } z \neq w. \text{ However,}$$

$$(g \circ f)(z) = (g \circ f)(w) \Rightarrow z = w$$

which is absurd.